Machine Learning for Imaging – Coursework Report

Mammographic density prediction

Group Leader: <**Sancaktaroglu**>

Ahmet Sancaktaroglu, Federico Mandelli
{ahmet.sancaktaroglu25, federico.mandelli25}@imperial.ac.uk

1 Part A: Choose a sensible data augmentation pipeline

Designed a data augmentation pipeline with both geometric and photometric transformations, chosen specifically for the mammography domain.

Geometric augmentations are applied to the single-channel image before conversion to three channels. We used random horizontal flipping (p=0.5) to make the model invariant to breast laterality (left vs right) and random affine transformations (p=0.5) with up to ±10° rotation, 5% translation and ±10% scaling to simulate minor patient positioning variations during acquisition. We deliberately excluded vertical flipping as mammograms have a consistent top-to-bottom anatomical orientation (chest wall at top).

Photometric augmentations are applied after converting to three channels to ensure compatibility with pretrained model expectations. We used random brightness and contrast jitter (±15%, p=0.5) to simulate exposure variation across different mammography machines and Gaussian blur (kernel size 3, p=0.3) to account for slight focus differences.

All augmentation parameters are kept mild to preserve the clinically meaningful tissue texture patterns that distinguish density classes. Additionally, we apply ImageNet normalisation (mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]) to all images to match the expected input distribution of our pretrained backbones. These augmentations increase the effective dataset diversity and help reduce overfitting by exposing the model to plausible variations in acquisition conditions.

2 Part B: Implement a sensible model for image classification

Evaluated three pretrained CNN architectures fine-tuned for 4-class breast density classification: **ResNet-18**, **DenseNet-121**, and **EfficientNet-B0**. All models were initialised with ImageNet weights and modified by replacing the original classification head with a dropout layer (p=0.5) followed by a linear layer mapping to 4 output classes. This transfer learning approach leverages low-level features (edges, textures) learned from ImageNet that transfer effectively to mammographic tissue patterns. Given the relatively small dataset size and the subtle texture-based classification task, transfer learning improves sample efficiency and provides a strong inductive bias for medical image analysis.

All models were trained for 10 epochs with Adam optimiser (lr=10^{-4}, weight_decay=10^{-5}), batch size 64, cross-entropy loss and a ReduceLROnPlateau scheduler monitoring validation AUC. Class imbalance was handled via the ImbalancedDatasetSampler.

Model	Val AUC	Test AUC	Best Epoch	Parameters
ResNet-18	0.924	0.916	4	11.2M
DenseNet-121	**0.932**	**0.920**	2	7.0M
EfficientNet-B0	0.925	0.919	4	4.0M

Table 1: Validation and test performance of evaluated models.

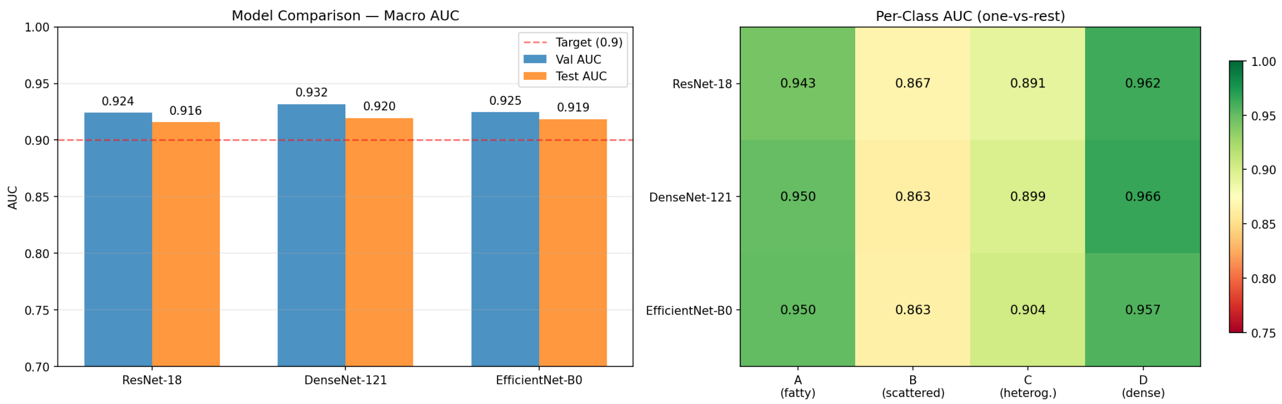

Figure 1: Overall model comparison: macro AUC on validation and test sets (left) and one-vs-rest per-class AUC heatmap (right).

Model selection was performed using validation AUC, while final performance is reported on the held-out test set to ensure unbiased evaluation. All three models exceed the 0.9 AUC target. **DenseNet-121 achieved the highest validation AUC (0.932)** and was selected for Parts C and D (Table 1). Its dense connectivity encourages feature reuse and improves gradient flow, which is particularly beneficial for capturing fine-grained texture variations between density classes.

Per-class analysis (Figure 1) reveals consistent patterns across all models: Classes A (fatty, ∼0.95) and D (dense, ∼0.96) are easiest to classify, while Class B (scattered, ∼0.86) is the most challenging. This is clinically expected, as the B/C boundary is the most subjective distinction even among radiologists.

Training curves (Figures 2–4) show some overfitting. Train AUC increases steadily while validation AUC plateaus around 0.92–0.93. The early best checkpoints (epoch 2–4) confirm that longer training does not improve generalisation, suggesting the augmentation and dropout regularisation are effective but the gap indicates room for further regularisation strategies. A limitation of this work is the limited exploration of hyperparameters and training duration, which could be investigated further in future work.

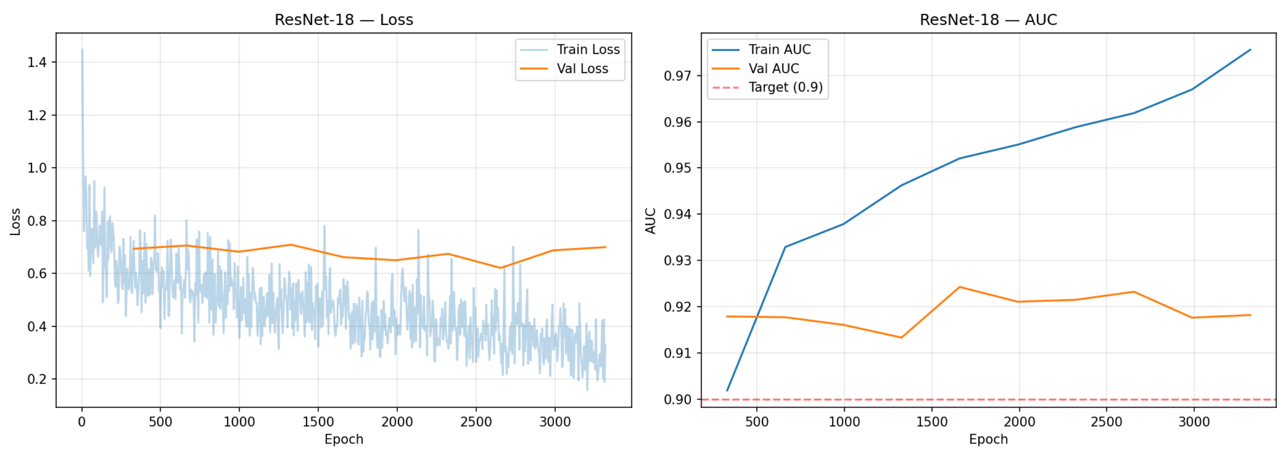

Figure 2: ResNet-18 training dynamics: loss (left) and AUC (right) across epochs.

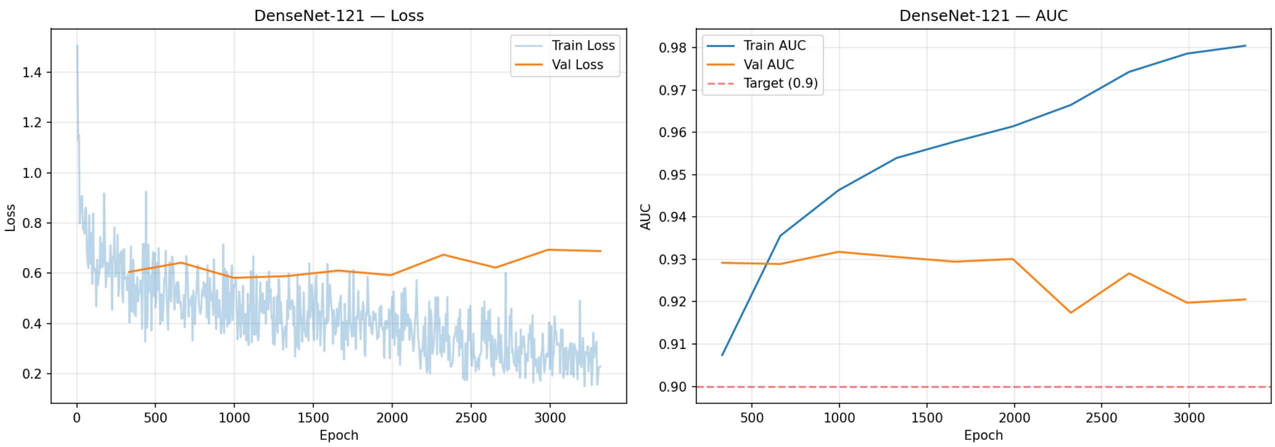

Figure 3: DenseNet-121 training dynamics: loss (left) and AUC (right) across epochs.

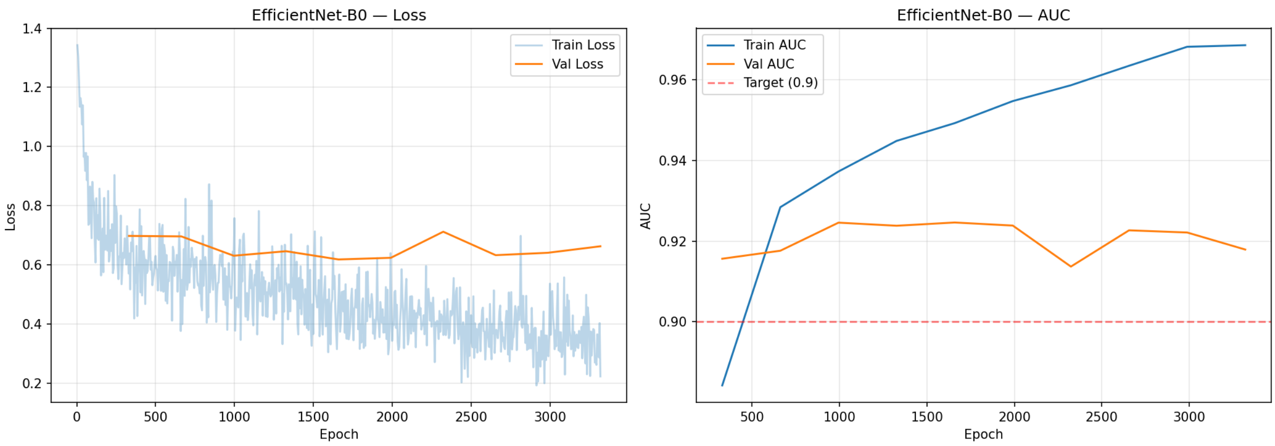

Figure 4: EfficientNet-B0 training dynamics: loss (left) and AUC (right) across epochs.

3 Part C: Conduct a subgroup performance analysis

The test set was stratified by clinically relevant metadata to assess subgroup performance. For each subgroup we report one-vs-rest AUROC per density class and the macro-AUROC; sample size is shown to indicate the statistical reliability of the estimate.

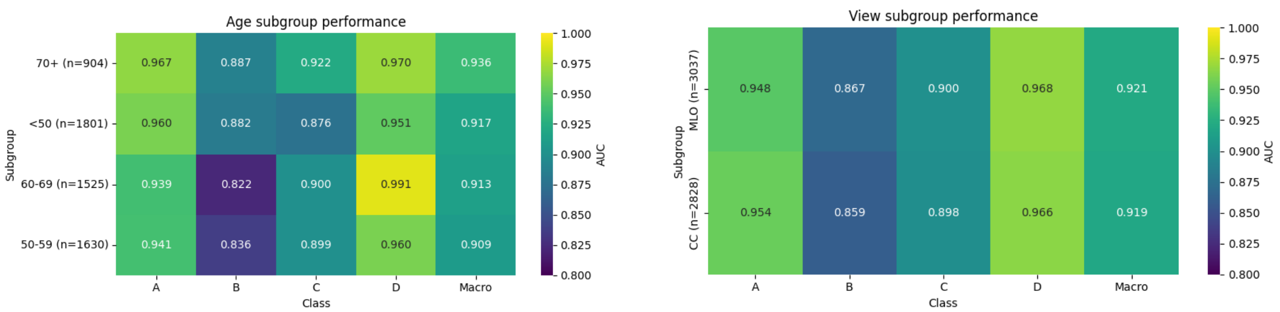

Figure 5: Subgroup performance without relevant disparity.

No clinically relevant disparity is observed across age or acquisition view. Age groups (<50, 50–59, 60–69, 70+) show stable performance (macro-AUROC 0.909–0.936), consistent with good generalisation across population distributions despite the known decrease in breast density with age. Similarly, CC and MLO views yield nearly identical results (0.919 vs 0.921), indicating robustness to changes in projection geometry and limited reliance on view-specific features. Class B remains the most challenging across all settings.

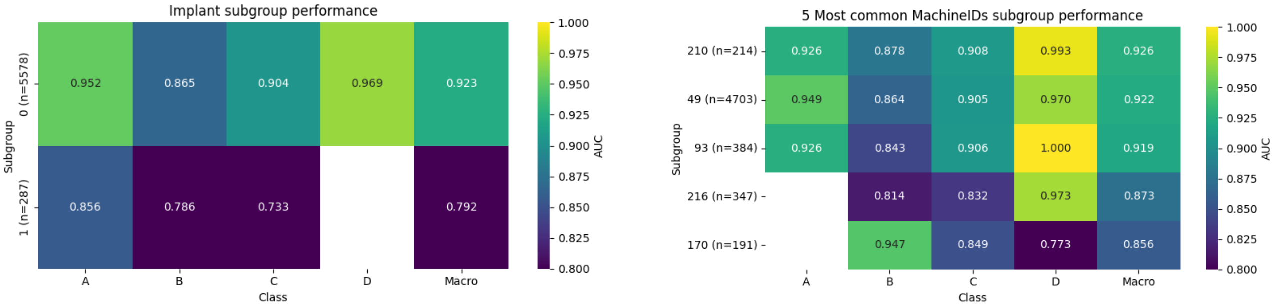

Figure 6: Subgroup performance showing clinically relevant disparities.

In contrast, patients with implants show a significant degradation (macro-AUROC 0.792 vs 0.923). This subgroup is substantially smaller, reducing the reliability of the estimate and suggesting under-representation during training. Implants also introduce consistent high-intensity regions and alter tissue visibility, resulting in a distribution shift relative to native breast anatomy which the model handles poorly.

Performance varies to a lesser extent across machines (macro-AUROC 0.856–0.926). Machines 216 and 170 show the largest degradation, with a pronounced drop for class D on machine 170 (AUC 0.773). This might indicate a scanner-dependent domain shift, likely due to differences in image contrast, resolution and vendor-specific post-processing.

4 Part D: Model inspection

Feature embeddings were projected using t-SNE to visualise local neighbourhood structure and detect subgroup-specific distributions, while PCA was used to assess whether these effects organise the feature space at a global scale.

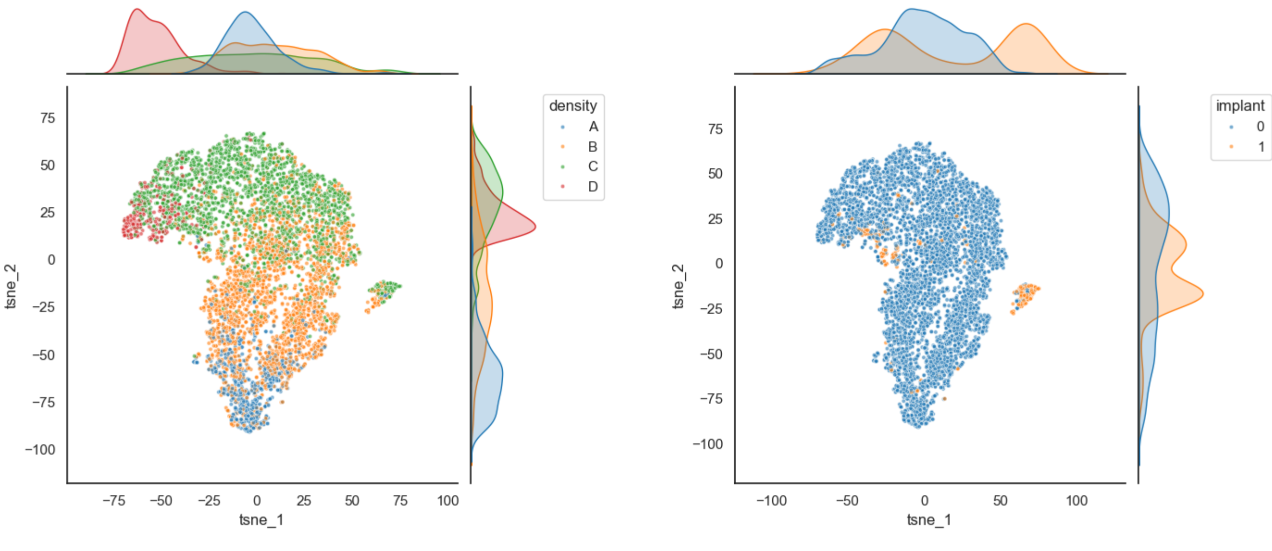

Figure 7: t-SNE projections coloured by variables that structure the embedding: density (left), implant (right)

When coloured by density, the t-SNE projection confirms the model has learned task-aligned features: classes A and D occupy opposite extremes and are well separated, while B and C lie in between and partially overlap. This reflects the fact that breast density varies gradually and is consistent with the lower separability of class B observed in Part C.

Implant cases are concentrated in a limited and shifted region of the embedding space and do not cover the full density range. This indicates that they form a different feature distribution, which aligns with the strong performance drop observed for this subgroup.

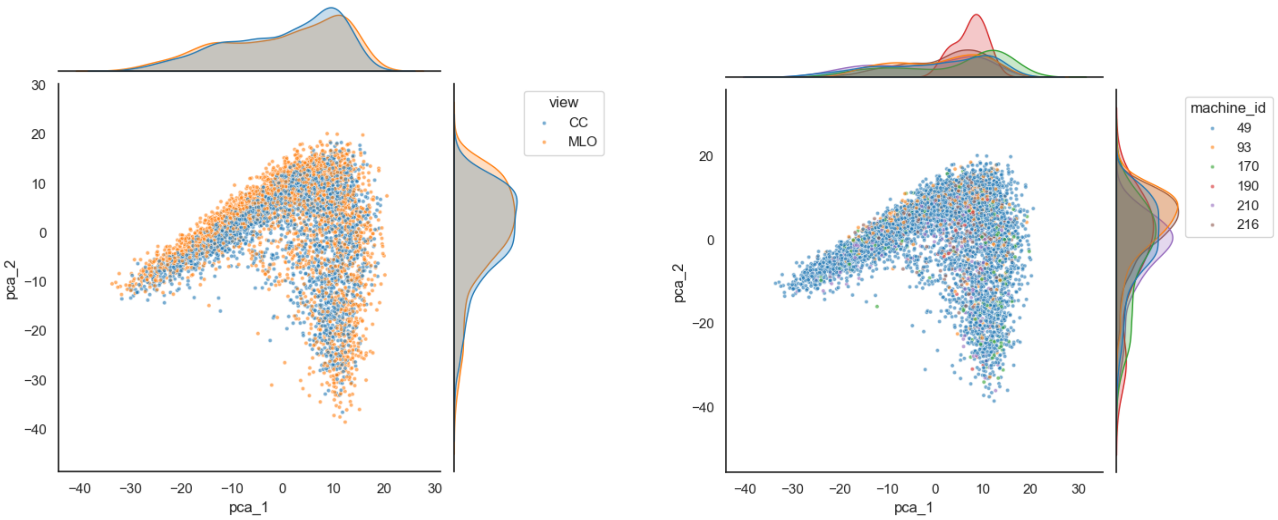

Figure 8: PCA projections coloured by variables that do not structure the embedding: view (left), machine ID (right)

Colouring by acquisition view shows strong overlap, indicating that the representation is largely view-invariant, which is consistent with the similar AUROC for the two views.

Samples from different machines are broadly intermixed with no clear scanner-specific regions, indicating that machine type does not organise the global feature space. However, subtle class-dependent shifts or differences in class distribution may still affect separability, which is consistent with the performance drop observed for specific scanners in Part C.